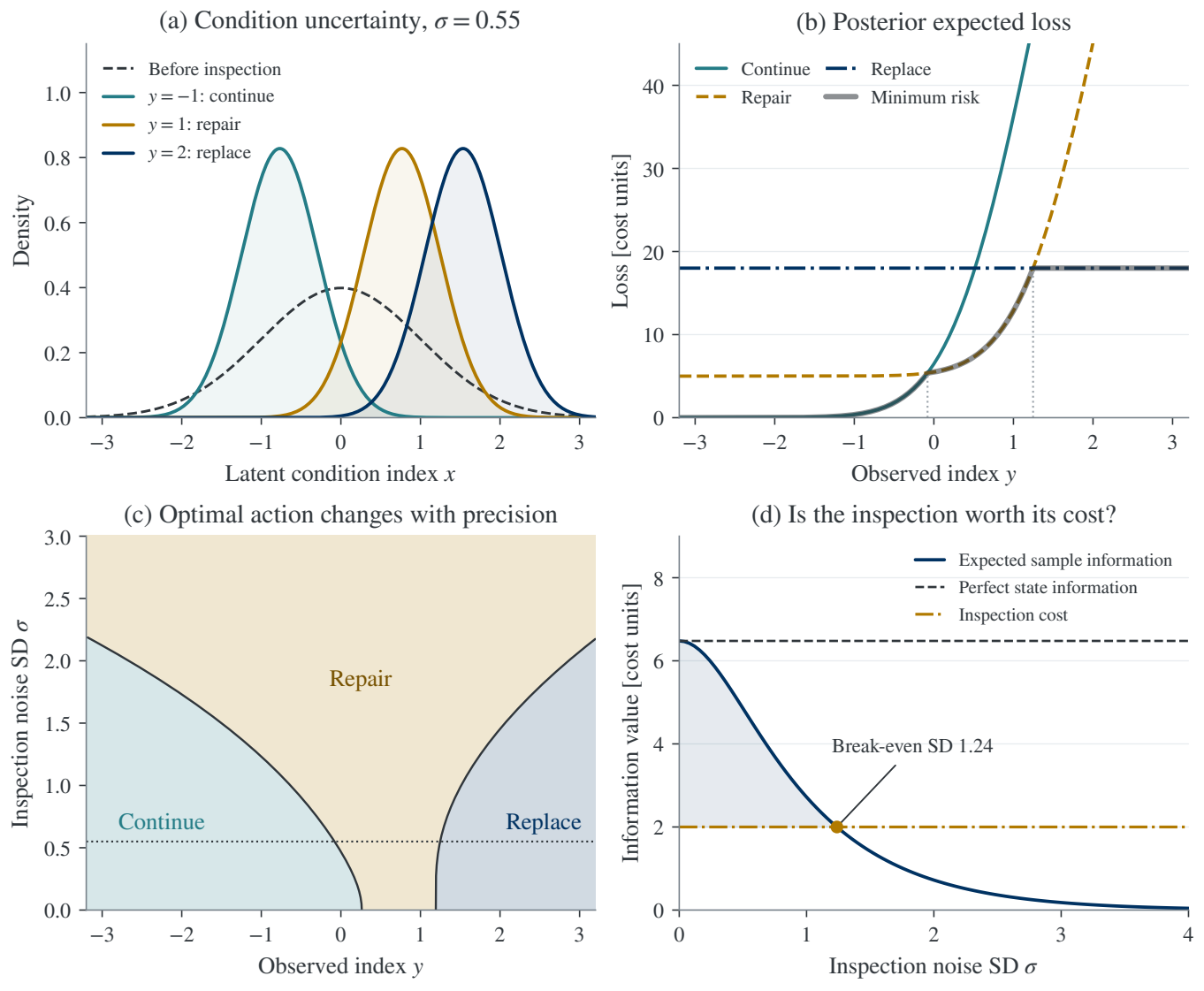
(a) Condition uncertainty, $\sigma = 0.55$
Before inspection
$y = -1$: continue
$y = 1$: repair
$y = 2$: replace
Density
Latent condition index x
(b) Posterior expected loss
Continue
Replace
Repair
Minimum risk
Loss [cost units]
Observed index y
(c) Optimal action changes with precision
Repair
Continue
Replace
Inspection noise SD σ
Observed index y
(d) Is the inspection worth its cost?
Expected sample information
Perfect state information
Inspection cost
Break-even SD 1.24
Information value [cost units]
Inspection noise SD σ